

AI Agent for Email Compliance Detection

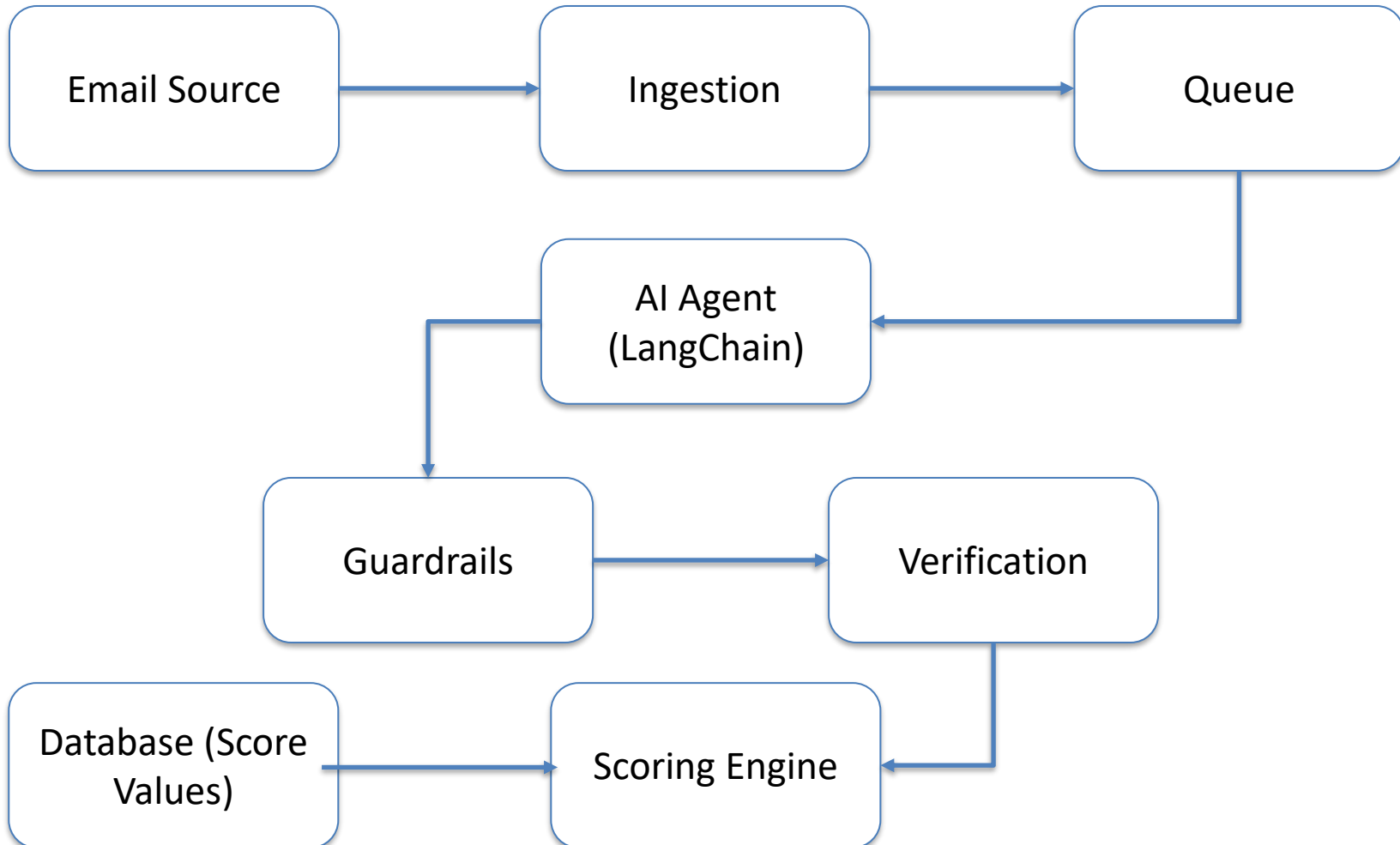
Problem Statement

- Manual email monitoring is inefficient
- Too many false positives
- No prioritization
- High manual effort

Solution Overview

- AI Agent analyzes emails
- Detects sender intent
- Assigns confidence score
- Prioritizes alerts automatically

System Architecture



Layered Architecture

Data Layer (Email Source, Kafka, Database)

AI Agent Layer (LangChain + Tools)

Validation Layer (Guardrails + Verification)

Scoring & Output Layer

Step 1: Ingestion

- Emails collected from source
- Pushed to Queue for scalable processing

Step 2: AI Agent

- LangChain agent reads email
- Understands intent of the sender
- Calls tools if needed
- Scores the email based on intent of the user.
- Generates structured output, preferably JSON.

Step 3: Guardrails

- Validate AI output
- Ensure correctness
- Prevent weak or missing reasoning

Step 4: Verification

- Confirms classification
- Reduces hallucination

Step 5: Scoring

- Assign priority score
- Based on category and severity
- Configurable via database

Step 6: Storage

- Store alerts in database
- Used for dashboards and review

Demo Flow

- Input: Email text
- Output: Category, Intent, Confidence, Evidence, Score
- System processes each email automatically

Key Benefits

- Accurate intent detection
- Reduced false positives
- Explainable results
- Scalable and configurable